

The keyboard

There is no Ctrl key: Fn stands in. It is sticky, so tap **Fn**, let go, then the key.

Fn C	stop what is running
Fn Z	pause it (fg goes on)
Fn 1 4	screen 1 to 4
Fn ←	the screen before, after
Fn 5 6	brightness down, up
Fn 7 8	volume down, up
Fn 9	sound off and on
Fn 0	doze: dark, all kept (page 7)
Fn ↓	look back at what scrolled away
Fn ↑	start, end of the line
Fn Del	delete forwards
Fn L	clear the screen
Esc	clear the line; leave a program
Tab	finish a command or a name
↓	what you typed before
Sym	the symbols printed on the keys
Shift ×2	capitals until pressed again

Brightness and volume are remembered for the next start. On a USB keyboard, Ctrl is Ctrl, and Ctrl-A then 1-4 changes screen.

Screens and the shell

Four screens, each with its own shell, all running at once: music on 1, a game on 2, notes on 3.

TAB DOES THE TYPING

theme **cu** **Tab** theme cursor
play "0a" **Tab** the whole name, quoted
video **Tab** only clips, and folders
When there are several it lists them; type more and Tab again.

LOOKING BACK

Fn **A** then **↑** scrolls back through what went past; **↑** **↓** go on, **Esc** comes back. The bar shows 122: that many lines back.

SEVERAL AT ONCE

play x.mp3 & runs behind; the shell is free
jobs what runs behind, or is paused
fg / bg to the front, or on behind
kill %1 stop job 1

PLUMBING

```
ls -l | sort -k5 -n | tail -3
grep -rn exam ~/notes > found.txt
find ~ -name '*.mp3' | wc -l
```

Power and battery

THREE WAYS TO REST

Fn **0** **doze**: dark and quiet, all kept; a key and you are back where you were
suspend **sleep**: any key wakes it, as a fresh start; alarms still wake it

poweroff deepest: only **Enter** wakes it
Left alone for 15 minutes with nothing running, it goes to sleep by itself (power suspend 30m to change it).

CHARGING

Plug the charger module's own USB-C in: 1 A, full in about three hours. The board's port runs the machine from the PC but does not charge.

HOW MUCH IS LEFT

DOING	LASTS, ABOUT
reading, screen at half	1 day
music, screen dark	30 hours
video or games	12 hours
dozing	3 days
asleep	4 days and more

battery -v tells all, battery -l the log of starts and sleeps.

First steps

WAKING IT

Press any key. After three minutes without one the screen dims, after four it goes dark: the next key only lights it again.

A COMMAND

Type one and press **Enter**. Try these:

- ls what is here
- cd music go into a folder; cd alone goes home
- play song.mp3 play it
- cal this month
- battery how much is left

THE BAR

1/4 18:34 73%

⚡ while it charges) · Wi-Fi

HELP

help every command there is
help play what one does, and how
man intro the manual, from the start
Tab finishes what you are typing

What it is

A computer that fits in a pocket, running a small Unix of its own. You talk to it by typing; it answers in text, on four screens at once.

It keeps your notes, a calendar, a to-do list and alarms; it plays music and video; it runs NES games; it reaches the internet over Wi-Fi; and it has its own programming language, pico, with its own bookie.

WHAT IS IN IT

- brain ESP32-S3: two cores at 240 MHz, 8 MB of memory
- screen 2.8", 320 × 240, 53 × 23 letters
- keys the CardKB, a keyboard the size of a card
- sound a speaker, and a microphone
- storage a microSD card; all your files
- power a 2500 mAh cell; days of use
- radio Wi-Fi, and Bluetooth for a gamepad

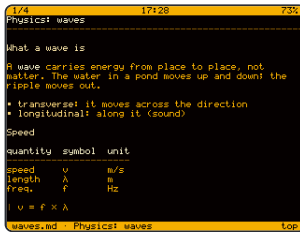
WHERE THINGS ARE

Your files are the SD card, which is your home, ~. Take it to a PC and they are at the top of it.

- ~/music ~/video songs and clips
- ~/notes ~/photos notes to read, pictures
- ~/roms ~/pico games, pico programs
- /tmp scratch space, empty after every start

Notes

Put Markdown or text files in ~/notes (a folder for each subject) and notes lists them; pick one to read it, laid out for the screen: headings, lists, tables, quotes.



Space **B** next page, back
0 the headings, to jump to one
/ **N** find a word, and the next
+ **-** brighter, darker, for reading in bed
E open it in the editor
Q close it; it reopens where you left it

WRITING

edit FILE: **Fn** **S** save, **Fn** **Q** quit, **Fn** **F** find, **Fn** **0** go to a line, **Fn** **K** cut, **Fn** **U** paste, **Esc** all of it in a menu.

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How to read it: commands look like this: type them and press **Enter**. **Fn** **X** means tap Fn, let go, then tap X. help and man on the machine say more about everything here.

PART 1 OF 2 · THE MACHINE

PocketType

the pocket guide



Part 1 getting started, keys, the shell, power and battery, notes
Part 2 calendar, music, video, games, the internet, looks, every command

1.0 beta 5

